KALAM (கககக) — AGRICULTURAL INSIGHTS ATLAS

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CROPPING INTENSITY

12.9% Double-Cropped Cropland

AUDIT READINESS

205 Targeted Audit Parcels (GPS)

ANNUAL WATER DEMAND

39.12 Million m³ (MCM) Canal Requirement

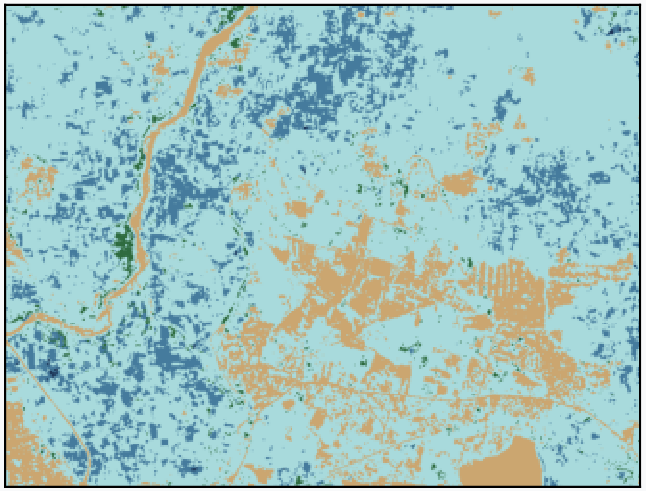
EXECUTIVE INSIGHT SUMMARY & POLICY ACTION ITEMS

- **Cropping Intensity:** 12.9% of cropland (370.6 ha) is double-cropped with Kar and Pishanam paddy cycles, with 1.4% under perennial banana.
- **Transplanting Calendar:** Peak Kar transplanting occurs in Jul (18.6% of fields), while main Pishanam sowing peaks in Nov. 0.0% of Pishanam acreage experiences delayed transplanting beyond November.
- **Field Verification Priority:** 25.0% of parcels (511 fields) are flagged for targeted ground audit due to lower phenological confidence or inter-annual crop instability, with top-50 priority targets packaged for GPS navigation.
- **Village-Level Statistics:** Kallidaikurichi leads overall agricultural cultivation across both crops with 614.4 ha of paddy (68% share) and 7.7 ha of banana.
- **Irrigation Dependency & Water Access:** 14.7% of paddy acreage is concentrated within 500 m of a river channel, canal or irrigation tank. Banana accounts for 1.4% of cropland in the immediate 0–250 m corridor, decaying sharply beyond 1 km.
- **Crop Change & Farmland Dynamics:** Evaluated 3,011.2 ha of 3,011.2 ha cropland (confidence ≥ 0.60): 52.9% of paddy and 1.3% of banana parcels remained stable across 2023–2026, with 100.5 ha (3.3%) identified as candidate paddy-to-banana diversification.
- **Crop Health & NDVI Anomaly:** 85.6% of analyzed cropland displays normal or above-normal vegetative vigour. Localized moisture or transplanting delay stress affects 14.4% (372.2 ha), predominantly along tail-end distributaries.
- **Indicative Gross Irrigation Requirement:** Total indicative gross irrigation water requirement is 39.1 MCM/year. Dual-season paddy accounts for 87.5% (34.2 MCM) of total demand, while perennial banana accounts for 2.1% (0.8 MCM).
- **Segment Size Distribution (Image-Derived):** 19.8% of image-derived field segments (9.9% of agricultural area) are below 1.0 ha. The mean segment size is 1.64 ha with a median of 1.41 ha.

Kalam © 2026 · Problem Statement 1 · SustainAI / Mapathon · Contains modified Copernicus Sentinel data

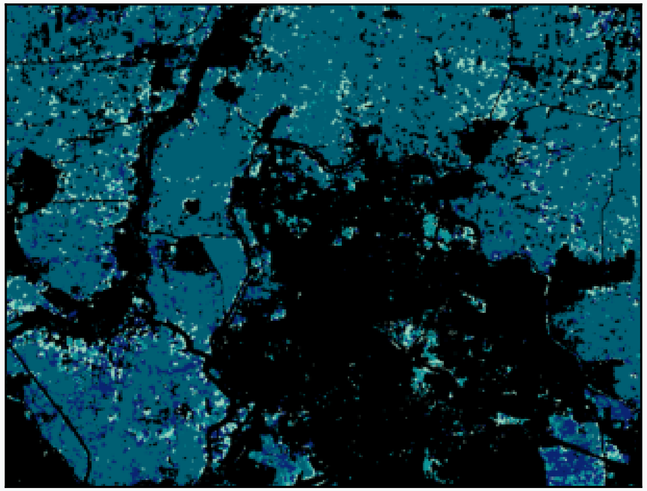
PHENOLOGY DYNAMICS & VERIFICATION PRIORITIZATION

01. Cropping Intensity



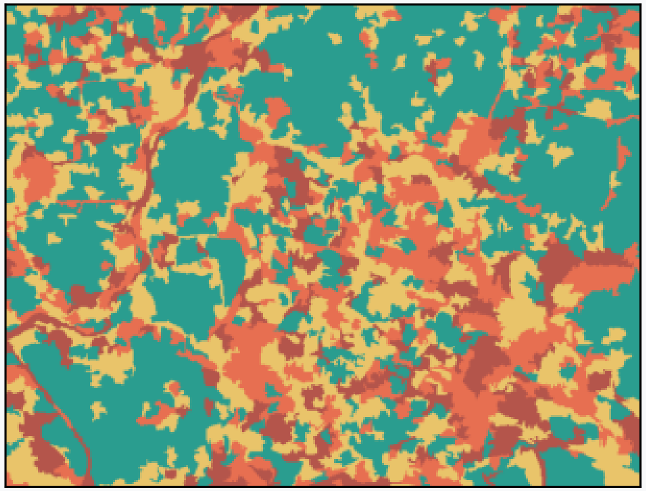
tivated (479.7 ha, 16.7%) | Single Crop (2453.6 ha, 85.6%) | Double Crop (370.6 ha, 12.9%) | Triple Crop (2.1 ha, 0.1%)

02. Transplanting Calendar



Kar Peak (Jun-Jul) | Kar Late (Aug) | Pishanam Peak (Oct) | Pishanam Delayed (Nov-Dec)

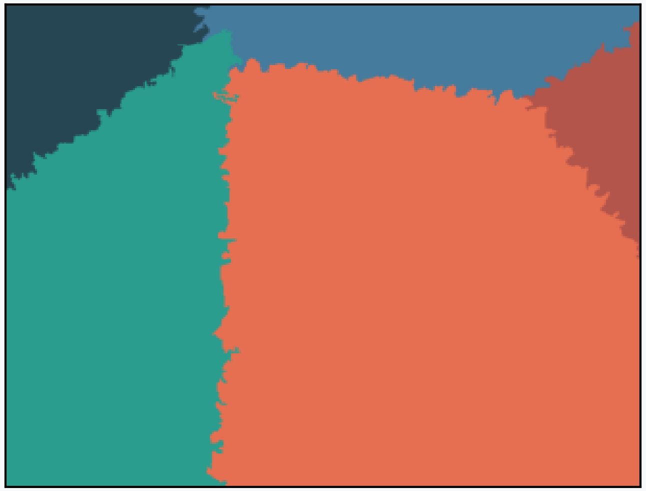
03. Field Audit Priority



Critical Priority (Top 10%) | High Priority (Next 15%) | Medium Priority (Next 25%) | Low Priority (Baseline 50

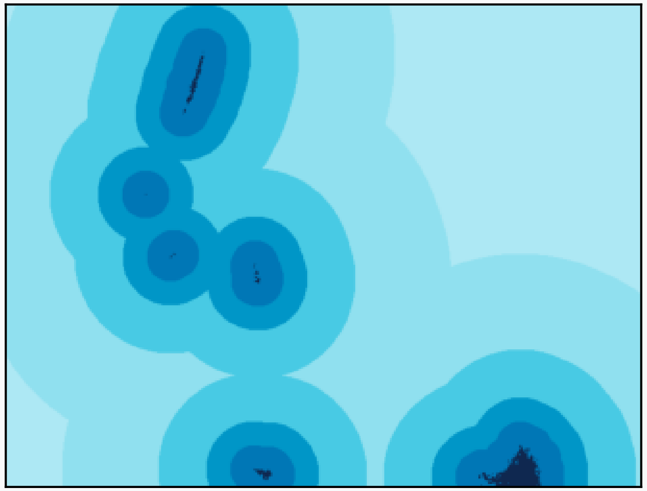
CANAL BUFFER DEPENDENCY & MULTI-YEAR CROP TRANSITIONS

04. Village Crop Allocations



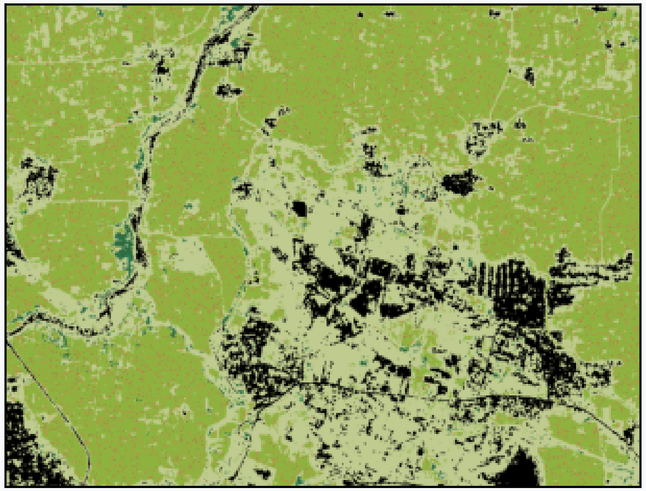
Urban (Paddy intensive) | Kallidaikurichi (Paddy intensive) | Veeravanallur (Diversified / Fallow) | Mukkudal (Paddy intensive)

05. Canal Buffer Decay



0 – 250 m | 250 – 500 m | 500 – 1,000 m | 1,000 – 2,000 m

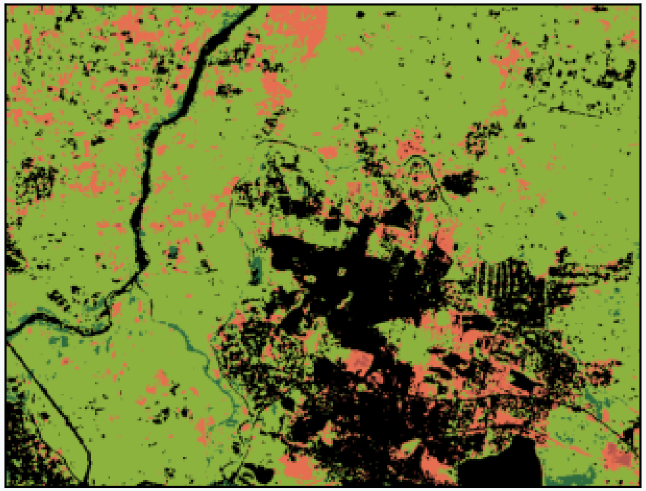
06. Crop Transitions (23–26)



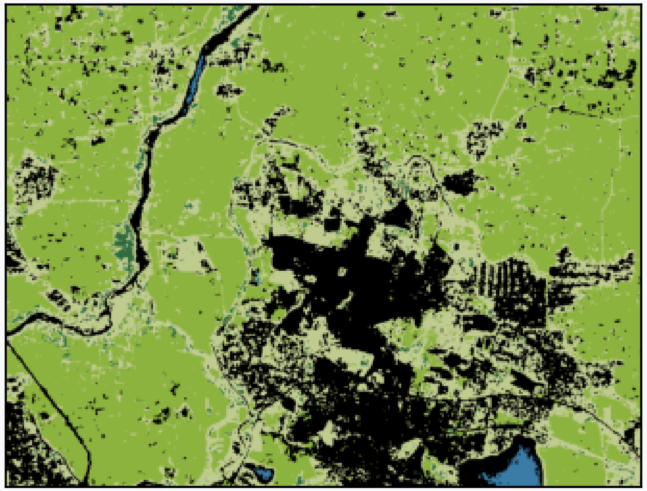
Stable Paddy | Stable Banana | Candidate Paddy → Banana | Banana → Paddy

CROP VIGOUR, WATER DEMAND & PARCEL MORPHOLOGY

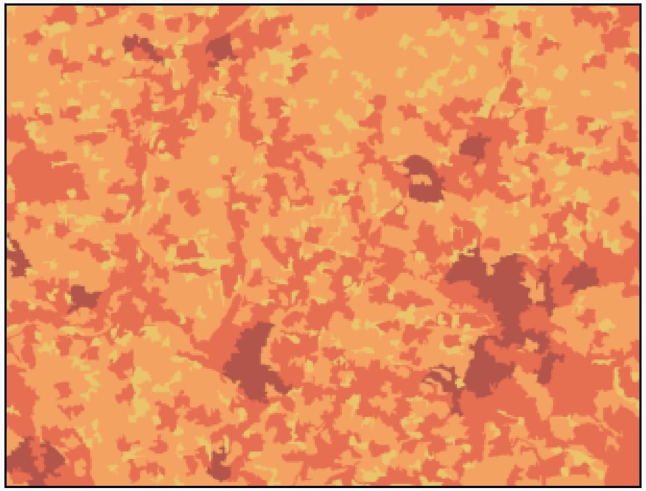
07. NDVI Vigour Anomaly



08. Water Demand Budget



09. Smallholder Fragmentation



ess (8.3 ha, 0.3%) | Mild Stress (363.9 ha, 14.1%) | Normal Vigou [illegible] nt) | Semi-Medium (1.0 – 2.0 ha) (50.6% area, 58.1% count) | Me